



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., V Semester & ECE A

Course Code & Title: EC8501& Digital Communication

Name of the Faculty member (s): Dr.S.Vairaprakash

Innovative Practice Description

- **Unit / Topic:** V/Channel Coding
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO14,TLO15
- **Activity Chosen:** Flipped Class room
- **Justification:**

The topic was chosen for flipped class because Channel coding covers error detection coding and error correction coding.. The Unit 5 deals Error Control Coding which covers Linear block codes ,cyclic codes, Syndrome calculation, convolution codes and viterbi decoder. Since the students are exposed to the digital logic operations, the topic of „error detection and control coding “ for flipped class was given . This topic can be learnt in self-learning mode with simple manipulations and logic. The students have chance to browse the topic to know how the code vectors are generated using matrix and schematic representations. And find more information related to error detection and correction.. Thus this flipped class activity is chosen to make them to work out different techniques to correct the errors.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The overview of flipped class activity was given to students before the implementation of the activity. The students were divided into groups and one topic is allotted to each group from Error detection/Correction topic as their wish. The topic related information and e-sources were given to the students. The students have to prepare a presentation for 10 minutes and each group has to present their topic chosen and others will listen to the presentation. After the presentation, question & answering is planned and others can share their ideas and doubts. By this way, all groups were shared their concepts learnt in their topic inside the class .

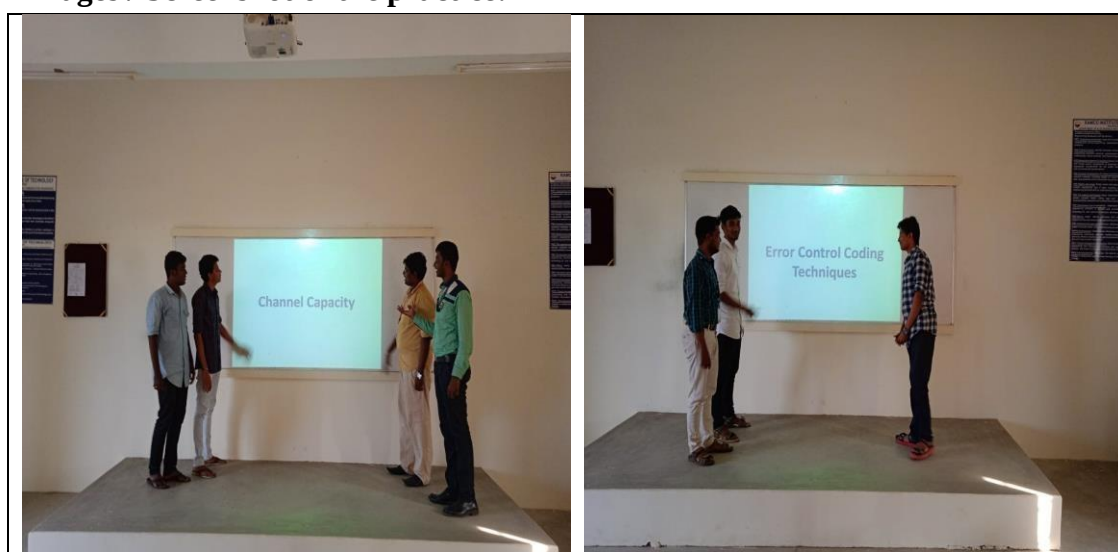
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO11	PSO1	PSO3
CO5	3	3	2	1	-	2	2	2

- **PO / PSO mapped:**

Innovative practice	PO3	PSO3
	2	2
Justification for correlation	The students will be able to develop coding methods are to bring good solutions for complex problems and to reduce errors; hence it is mapped to level 3.	The students will be able to design the channel encoder and decoder circuit in the communication system, hence it is mapped to level 2.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The students felt a new practice and it helped them to know to make good presentation and avoid stage fear. They liked this practice of teaching that can be done through this type of activity. They could get idea from learning sources which would help them to prepare the presentation.

- ❖ **Benefit of the practice:**

Students done this activity in a successful manner. They planned the activity for their chosen topic and prepared the presentation in the conceptual manner. All members in the group were learnt the topics thoroughly. By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each member in the group was able to present and deliver the concept of error correction and detection coding.

- ❖ **Challenges faced in implementation:**

Some of the students felt difficult to present because of lack of communication. Few groups done their presentation beyond their time limit. Few students in the group felt difficult to answer all the questions.

References:

1. B.Sklar, "Digital Communication Fundamentals and Applications", 2nd Edition, Pearson Education, 2009.
2. B.P.Lathi, "Modern Digital and Analog Communication Systems" 3rd Edition, Oxford University Press 2007.
3. H P Hsu, Schaum Outline Series - —Analog and Digital Communications, TMH 2006.
4. J.G.Proakis, "Digital Communication", 4th Edition, Tata McGraw Hill Company, 2001.



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Course Code & Title: EC8501& Digital Communication

Name of the Faculty member (s): Dr.S.Vairaprakash

Innovative Practice Description

- **Unit / Topic:** IV/ BER comparison of Digital Modulation Schemes
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO12
- **Activity Chosen:** Jigsaw
- **Justification:**

The activity „JIGSAW“ is used to recollect the concepts knowledgeable and to enrich the learning through this teamwork. The concept of generation, detection and error probability of various digital modulation schemes were already learnt in modulation techniques. By recollecting all the information in various modulation schemes, students from expert group were able to share the information to home group.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The students are divided into Home Groups with each group having 6 members . Each member in home group will form different groups named Expert group. Each expert group is allotted with each modulation technique The expert group members discuss the topic in detail for 20 minutes with the learning materials and e-sources

The Expert group members should go to their original home group and discuss the points (25 Minutes) to the other members. All the members in home group learnt about all the topics through expert group members.

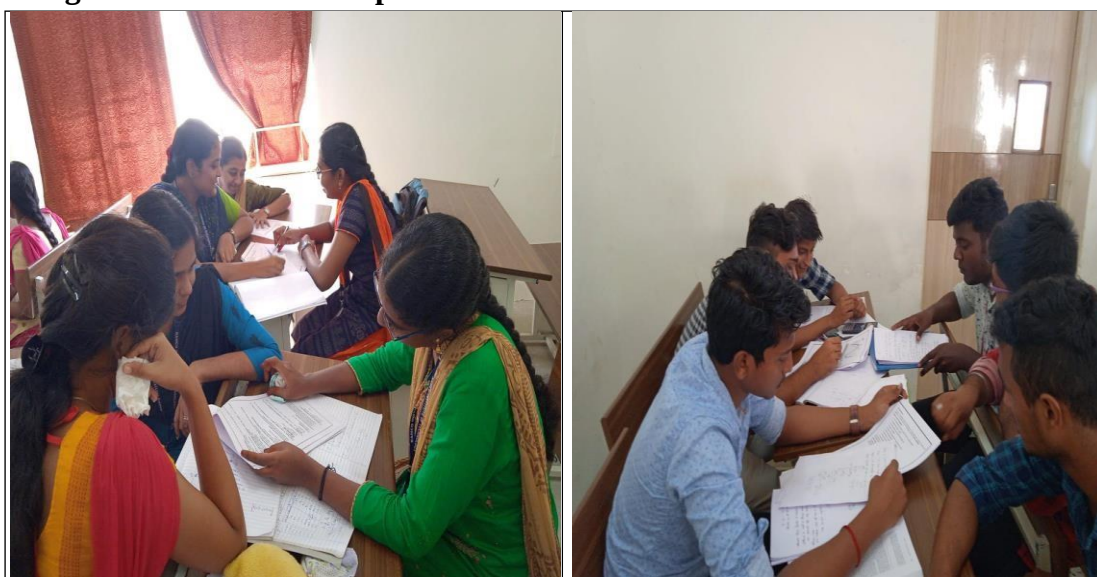
CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PSO1	PSO3	PSO4
CO5	3	3	2	2	2	1	3	3	2

- **PO / PSO mapped:**

Innovative practice	PO1	PSO1
	3	3
Justification for correlation	The outcome is mapped to level 3 because Modulation and demodulation concepts are essential to understand the digital modulation techniques.	The students will be able to understand the band pass transmission schemes for designing the special antennas and the underlying circuitry in the area of communication, hence it is highly mapped.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The students felt a new experience and they could able to recollect the information about digital modulation techniques. It helped them to learn the concept through collaborative learning with their class mates. They liked this practice of learning through this type of activity for recollecting the information that they already learnt.

- ❖ **Benefit of the practice:**

By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each Home group could recollect the information of seven modulation techniques .

- ❖ **Challenges faced in implementation:**

Initially, I have planned the activity for 45 minutes but some of the students takes more time to share the information.

Some students share minimal information to their home group.

References:

1. B.Sklar, "Digital Communication Fundamentals and Applications", 2nd Edition, Pearson Education, 2009.
2. B.P.Lathi, "Modern Digital and Analog Communication Systems" 3rd Edition, Oxford University Press 2007.
3. H P Hsu, Schaum Outline Series - —Analog and Digital Communications, TMH 2006.
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Innovative Teaching Method

UNIT I –Information Theory

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: Information and Entropy

Name of the Innovative Practice: One-minute Paper

Date & Duration: 02.09.2021 & 5 minutes

Sample Images:

Innovative Teaching Method Execution

Information and Entropy



I = S. Gomathir Naayagam
IIIrd YEAR ECE - A

Information Rate $R = \gamma H$

Information $I = \log_2 P_i$

Entropy $H = \sum P_i \log_2 \frac{1}{P_i}$

Where,

$P_i \Rightarrow$ Probability of symbols.



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Innovative Teaching Method

UNIT I –Information Theory

Degree, Semester& Branch: V Semester B.E. ECE A
Course Code & Title: EC8501 Digital Communication
Name of the Faculty member: Dr.S.Vairaprakash
Name of the Topic: Channel Capacity
Name of the Innovative Practice: Quiz
Date & Duration: 04.10.2021 & 15 minutes

Innovative Teaching Method Execution

Channel Capacity





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Innovative Teaching Method

UNIT I –Information Theory

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: Huffman Coding

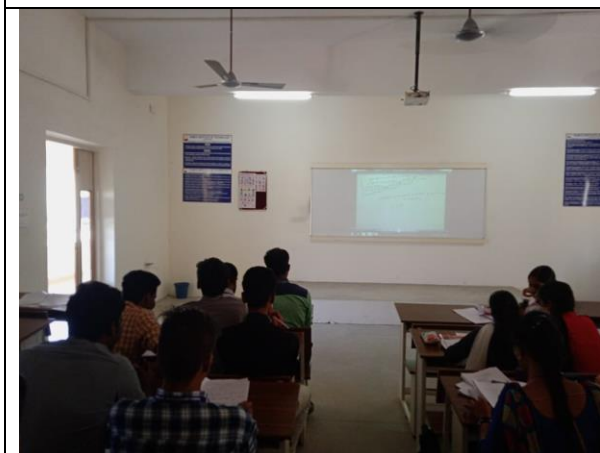
Name of the Innovative Practice: NPTEL Video

Date & Duration: 06.09.2021 & 10 minutes

Sample Images:

Innovative Teaching Method Execution

Huffman Coding





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Innovative Teaching Method

UNIT II –Waveform coding &Representation

Degree, Semester& Branch: V Semester B.E. ECE A
Course Code & Title: EC8501 Digital Communication
Name of the Faculty member: Dr.S.Vairaprakash
Name of the Topic: Properties of Line codes
Name of the Innovative Practice: Role play
Date & Duration: 20.09.2021 & 5 minutes

Innovative Teaching Method Execution

Properties of Line codes





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Innovative Teaching Method

UNIT II –Waveform coding &Representation

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: Bipolar NRZ and Manchester coding

Name of the Innovative Practice: Zero Minute Speech

Date & Duration: 01.10.2021 & 5 minutes

Innovative Teaching Method Execution

Bipolar NRZ and Manchester coding





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Innovative Teaching Method

UNIT III –Baseband Transmission & Reception

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: Pulse Shaping

Name of the Innovative Practice: Zero Minute Speech

Date & Duration: 09.10.2021 & 5 minutes

Innovative Teaching Method Execution

Pulse Shaping





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Innovative Teaching Method

UNIT III – Baseband Transmission & Reception

Degree, Semester& Branch: V Semester B.E. ECE A
Course Code & Title: EC8501 Digital Communication
Name of the Faculty member: Dr.S.Vairaprakash
Name of the Topic: Eye Pattern, Receiving Filters
Name of the Innovative Practice: Think Pair Share
Date & Duration: 18.10.2021 & 5 minutes

Innovative Teaching Method Execution

Eye Pattern, Receiving Filters





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Innovative Teaching Method

UNIT IV –Digital Modulation Scheme

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Dr.S.Vairaprakash

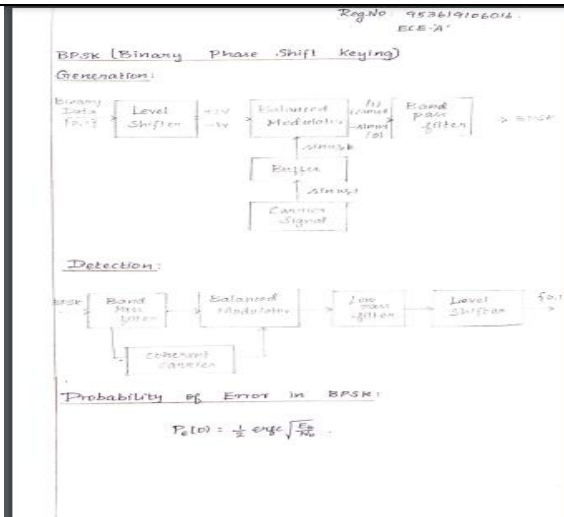
Name of the Topic: Generation, Detection, PSD &BER of Coherent BPSK

Name of the Innovative Practice: Sketch Noting

Date & Duration: 28.10.2021 & 45 minutes

Innovative Teaching Method Execution

Generation, Detection, PSD &BER of Coherent BPSK

 POCO SHOT ON POCO M2	 Page No. 9999999999 ECE-A BPSK (Binary Phase Shift Keying) <u>Generation:</u> Binary Data (0,1) → Level Shifter → +ve/-ve → Balanced Modulator → Carrier Signal → Band Pass Filter → BPSK <u>Detection:</u> BPSK → Band Pass Filter → Balanced Modulator → Low Pass Filter → Level Shifter → 0,1 <u>Probability of Error in BPSK:</u> $P_e = \frac{1}{2} \exp\left(-\frac{E_b}{N_0}\right)$
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Innovative Teaching Method

UNIT IV –Digital Modulation Scheme

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: Generation, Detection, PSD &BER of Coherent BFSK

Name of the Innovative Practice: Open Book Test (Library)

Date & Duration: 08.11.2021 & 45 minutes

Innovative Teaching Method Execution

Generation, Detection, PSD &BER of Coherent BFSK





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UNIT IV –Digital Modulation Scheme

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: Quadrature Amplitude Modulation

Name of the Innovative Practice: NPTEL Video

Date & Duration: 12.11.2021 & 10 minutes

Innovative Teaching Method Execution

Quadrature Amplitude Modulation

